

CREE LAKE, SK, Canada

WMO: 729200

Lat: 57.350N

Lon: 107.130W

Elev: 1623

StdP: 13.85

Time Zone: -6.00 (W06)

Period: 86-96

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -41.3	(c) -36.3	(d) -40.3	(e) 0.6	(f) -34.2	(g) -36.9	(h) 0.7	(i) -31.5	(j) 20.0	(k) 15.3	(l) 17.5	(m) 9.9	(n) 2.8	(o) 260	(p) 0.718

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 16.5	(c) 80.5	(d) 61.7	(e) 76.5	(f) 59.8	(g) 72.6	(h) 57.8	(i) 63.9	(j) 76.6	(k) 61.6	(l) 72.9	(m) 59.7	(n) 69.4	(o) 8.8	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.5	80.5	67.2	57.4	74.7	64.4	55.5	69.6	63.0	30.0	76.8	28.3	73.3	27.0	69.6	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.3	16.1	14.2	-47.6	87.0	6.6	4.0	-52.3	89.9	-56.2	92.2	-59.9	94.4	-64.7	97.3		
			-40.0	67.0	6.3	2.9	-44.6	69.1	-48.3	70.8	-51.8	72.4	-56.4	74.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	28.7	-5.0	-1.0	12.1	28.9	44.4	56.2	60.6	58.2	47.4	32.0	10.2	-1.8	(6)
(7)		DBStd	26.89	16.58	17.37	15.62	11.15	9.30	6.72	5.71	7.50	7.75	11.29	13.40	18.31	(7)
(8)		HDD50	8709	1705	1428	1175	633	229	23	0	12	141	561	1195	1607	(8)
(9)		HDD65	13321	2170	1848	1640	1083	642	269	165	237	528	1022	1645	2072	(9)
(10)		CDD50	930	0	0	0	1	54	210	329	267	64	5	0	0	(10)
(11)		CDD65	62	0	0	0	0	2	5	29	26	0	0	0	0	(11)
(12)		CDH74	615	0	0	0	0	30	67	223	290	5	0	0	0	(12)
(13)		CDH80	127	0	0	0	0	4	4	42	77	0	0	0	0	(13)
(14)	Wind	WSAvg	6.5	5.8	6.0	6.6	6.7	6.6	6.8	6.3	6.6	7.6	7.7	6.4	5.4	(14)
(15)	Precipitation	PrecAvg	18.7	0.7	0.6	0.7	0.9	1.2	2.6	3.7	2.8	2.0	1.5	1.0	0.9	(15)
(16)		PrecMax	24.6	1.4	1.5	1.7	2.3	2.7	5.6	6.4	4.4	4.4	3.4	2.4	1.5	(16)
(17)		PrecMin	13.8	0.2	0.1	0.2	0.2	0.1	0.8	1.7	0.9	0.7	0.4	0.3	0.3	(17)
(18)		PrecStd	3.0	0.3	0.4	0.4	0.5	0.7	1.3	1.5	1.0	1.1	0.7	0.6	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.6	44.2	46.2	59.9	78.6	80.0	84.8	87.4	73.0	64.0	37.6	34.6	(19)
MCWB			34.2	36.8	36.8	45.5	56.9	59.2	63.8	67.3	58.1	50.5	34.3	32.3	(20)	
2%		DB	30.7	36.5	42.2	52.3	69.9	75.7	80.1	81.1	67.6	57.0	33.2	30.2	(21)	
		MCWB	29.1	32.3	34.9	41.7	53.6	56.9	62.5	62.7	54.6	46.9	31.5	28.6	(22)	
5%		DB	24.0	30.3	38.3	47.9	64.9	72.3	75.6	76.7	64.0	51.1	30.9	26.0	(23)	
		MCWB	22.8	27.7	33.3	39.0	50.9	55.5	60.3	61.0	53.0	43.8	29.8	24.5	(24)	
10%		DB	18.4	24.7	34.1	43.9	60.5	68.7	71.8	71.5	59.6	46.2	27.8	21.9	(25)	
		MCWB	17.4	23.0	31.2	36.2	48.3	54.5	58.6	59.3	51.0	41.2	26.7	20.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.6	36.7	37.8	46.1	58.0	61.9	66.4	68.7	58.6	51.9	34.7	32.7	(27)
MCD B			38.6	42.8	44.7	57.1	74.1	71.2	75.8	83.9	70.6	62.9	36.7	34.6	(28)	
2%		WB	29.9	32.3	35.7	42.2	54.7	59.6	64.0	64.6	56.3	47.4	32.0	28.7	(29)	
		MCD B	31.7	36.2	41.1	51.1	67.9	68.9	76.3	77.5	64.9	55.4	33.1	30.0	(30)	
5%		WB	23.2	28.0	33.5	39.5	52.0	58.1	61.9	62.0	54.3	44.4	29.7	24.8	(31)	
		MCD B	24.3	30.5	37.8	47.0	63.3	67.6	73.1	73.7	61.8	50.8	31.0	26.4	(32)	
10%		WB	17.8	23.3	31.2	36.9	49.3	56.5	60.0	59.9	51.9	41.6	26.8	21.1	(33)	
		MCD B	18.6	25.0	34.5	43.0	59.2	66.2	69.2	70.2	58.6	46.0	27.8	22.2	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	19.1	20.6	22.9	19.0	19.7	18.0	16.5	17.4	14.9	12.2	13.1	16.1	(35)
MCDBR			23.0	20.9	23.2	22.6	26.1	23.7	21.7	25.0	21.3	20.4	14.0	16.1	(36)	
MCWBR			21.9	17.9	17.1	14.1	13.4	10.0	8.6	10.0	10.8	12.4	12.9	15.3	(37)	
5% WB		MCDBR	23.4	20.2	22.6	21.3	24.2	18.9	19.2	22.8	19.3	19.4	13.7	15.3	(38)	
		MCWBR	22.4	17.3	17.4	13.8	12.9	10.1	8.7	9.7	10.6	12.4	12.7	14.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.226	0.257	0.260	0.309	0.324	0.359	0.385	0.344	0.303	0.278	0.238	0.205	(40)	
taud		2.202	2.236	2.314	2.331	2.475	2.395	2.358	2.456	2.595	2.560	2.287	2.152	(41)		
Ebn at Noon		214	255	289	288	290	279	268	274	273	252	208	188	(42)		
Edh at Noon		15	23	29	34	31	35	35	30	22	17	14	12	(43)		
(44)	All-Sky Solar Radiation	RadAvg	212	531	1068	1582	1784	1832	1705	1363	897	462	186	133	(44)	
(45)		RadStd	17	31	44	92	200	147	106	106	113	68	24	15	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page